

Digital Logic Design

(EE1005)

Date: 1st March, 2025

Course Instructor(s)

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Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 3

Roll No. _____

Student Signature _____

- Attempt all the questions.
- Show complete working of each question.
- Multiple solutions of the same question will carry zero credit.
- State your valid assumptions clearly if you have to take any.

CLO #1: Describe various number systems and perform arithmetic operations and base conversions.

Q1: [15 marks]

- Convert the decimal number $(-73)_{10}$ into 2's complement 8-bit number
- Find the decimal equivalent of $(113.1)_5$
- Find the unknown in the following expression.
 $(2A4)_{16} + (111\ 1010\ 1101)_2 = (?)_{16}$
(Perform addition in binary and get result in hexadecimal format. Show complete working)

CLO #2: Apply Boolean Algebra and K-map methods to optimize logic circuits

Q2: [15 marks]

- Apply Boolean algebra to simplify the following function to minimum number of literals

$$F(W, X, Y, Z) = WXY + \overline{X + Y + Z} + \overline{WXZ} + \overline{WXY}$$

- Construct a simplified logic diagram of the following expression using 2-input NAND gates only. Inputs are available in true form only.

$$F(A, B, C) = A\overline{B}C + A\overline{C} + AB\overline{C}$$

CLO #2: Apply Boolean Algebra and K-map methods to optimize logic circuits

Q3:

[15marks]

- a) Design a magnitude comparator that compares two 2-bit numbers. The numbers are $A = (A_1A_0)$ and $B = (B_1B_0)$. The comparator has Three outputs G, L and E with $G = 1$ if $A > B$ and $L = 1$ if $A < B$ and $E = 1$ if $A = B$.
- Construct the truth table of the system.
 - Write the Boolean function representation of all the outputs, using "little m" notation?
 - Write the Boolean function representation of the output G only, using "big M" notation?
- b) Use K- map to find minimal POS (Product of sum) expression for the following Boolean functions

$$F(A, B, C, D) = \sum m(0, 2, 5, 8, 9, 10, 11, 12, 13)$$